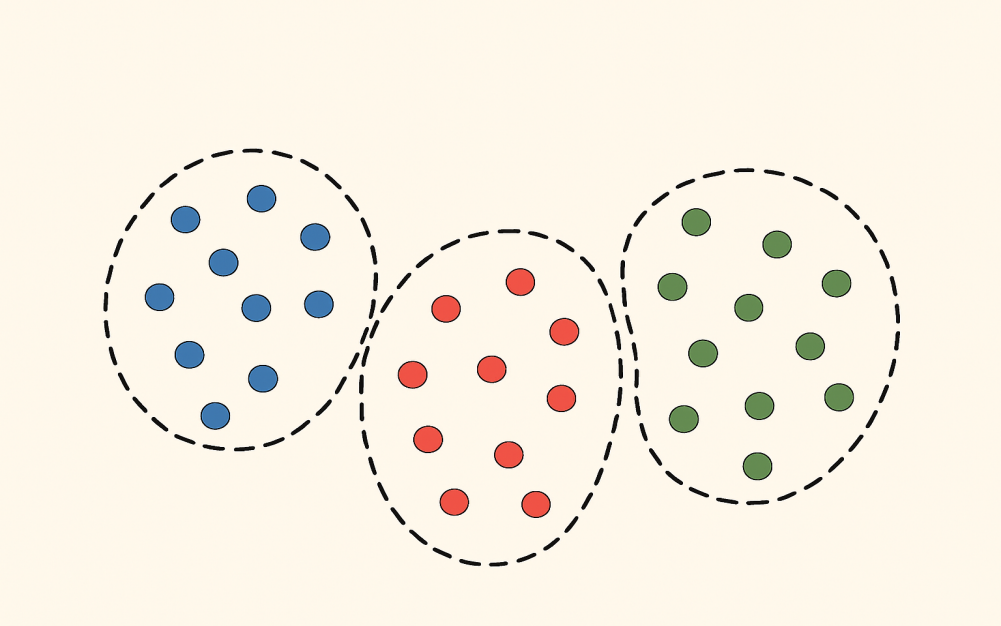
Dynamic k-Center Clustering

Antonis Skarlatos



Introduction

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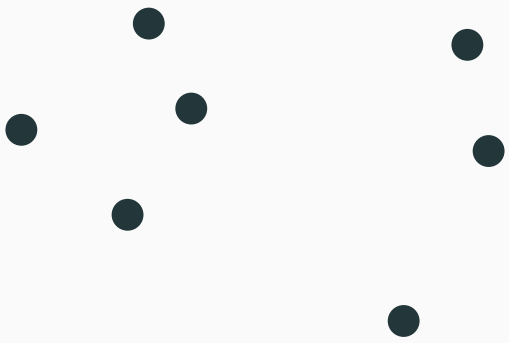

Definition (k-Center)

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Output: Subset $S \subseteq P$:

$|S| \leq k$

$\max_{p \in P} \mathrm{dist}(p, S)$ is minimized

k-Center Clustering Problem

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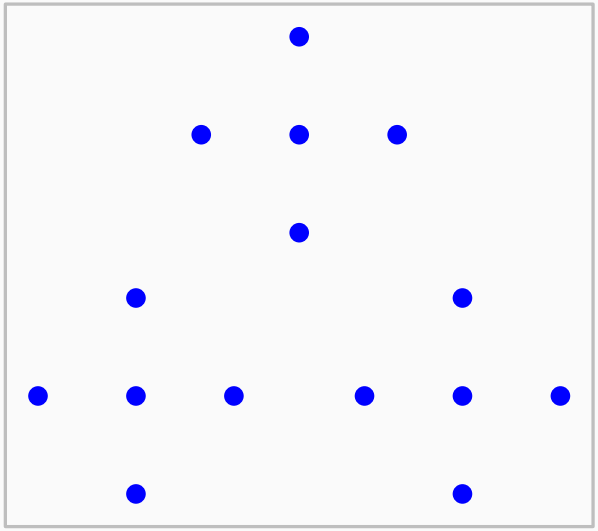
$k = 3$



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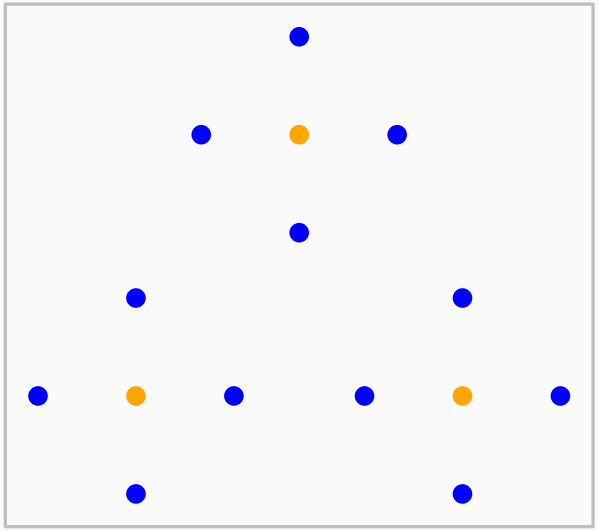

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Arbitrary Metric Space $\Rightarrow$
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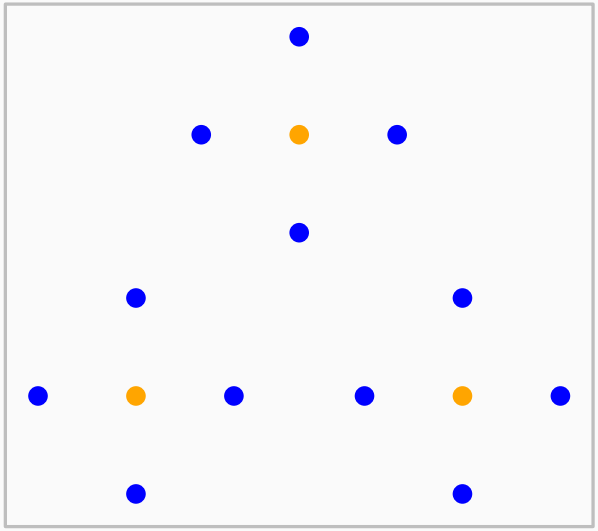

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$\Rightarrow$ **clusters**

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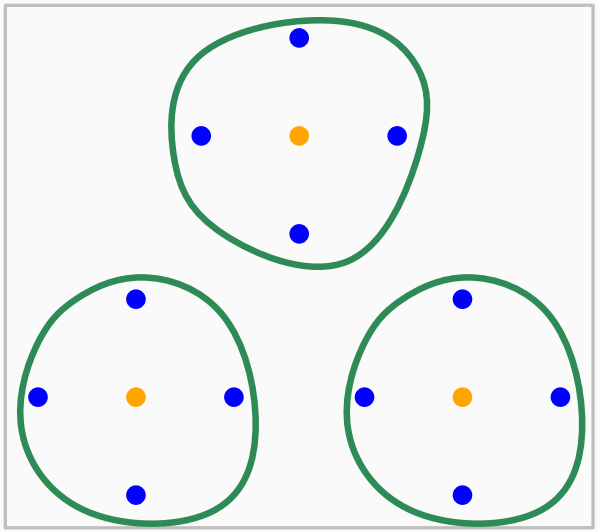

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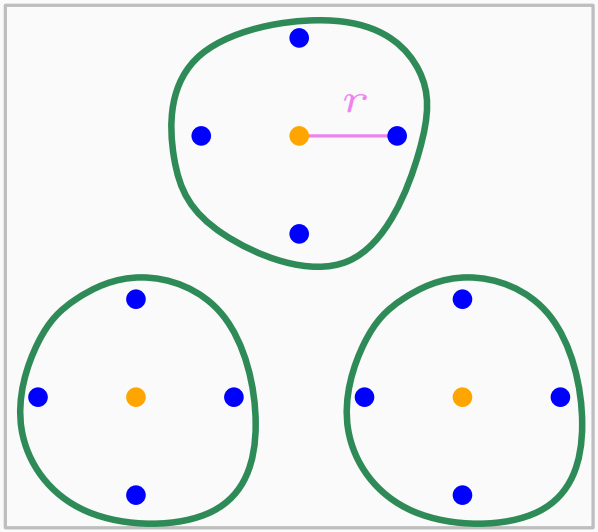

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NP-hard to approximate within a factor of $2 - \epsilon$

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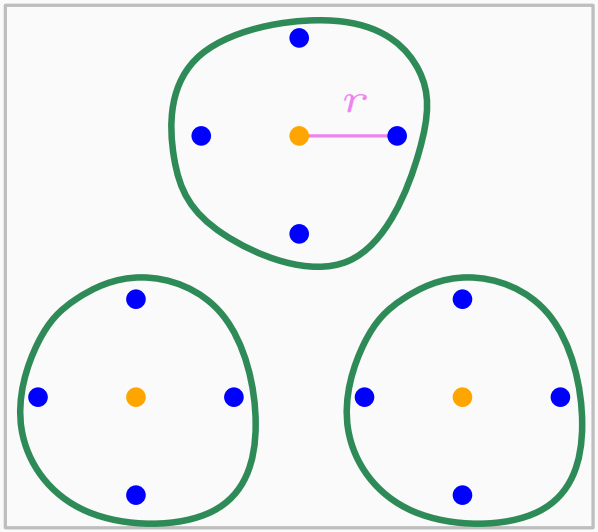
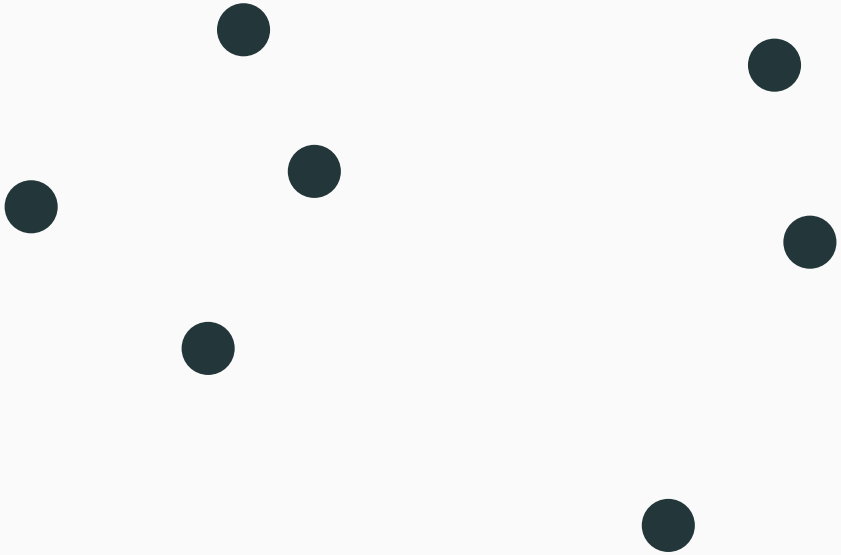


Dynamic Input



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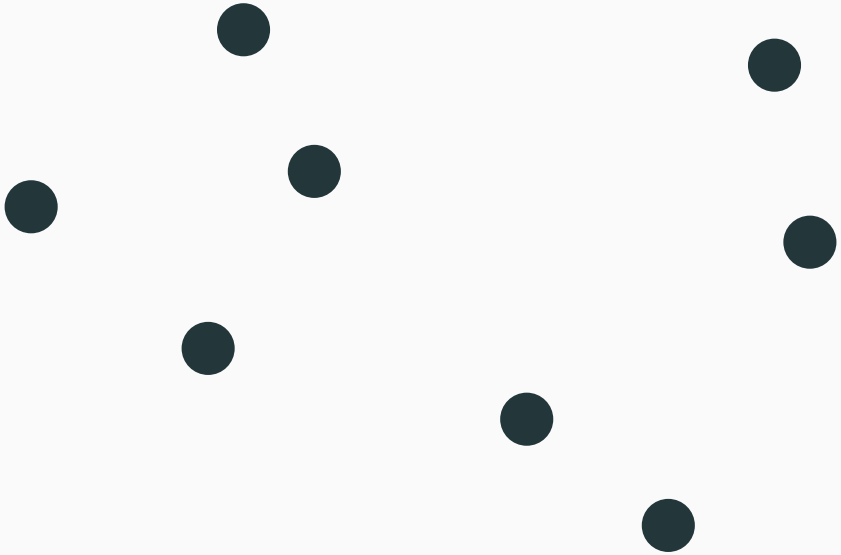

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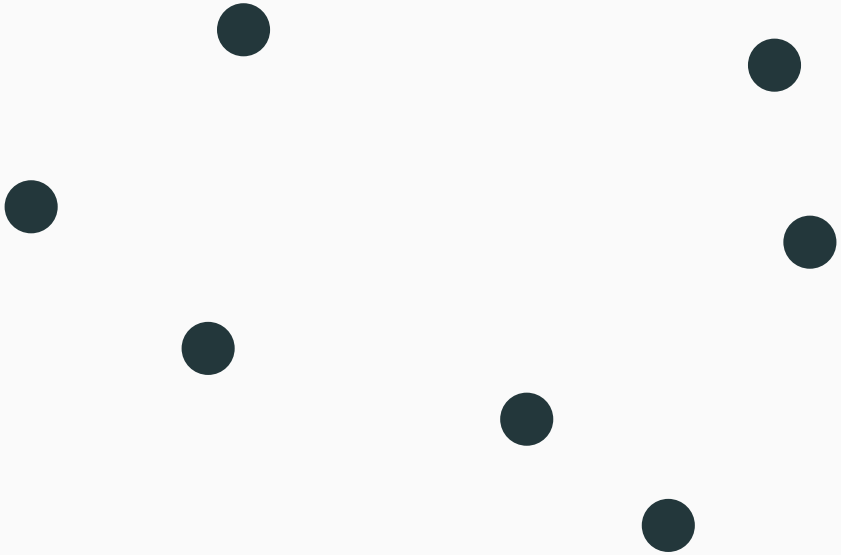

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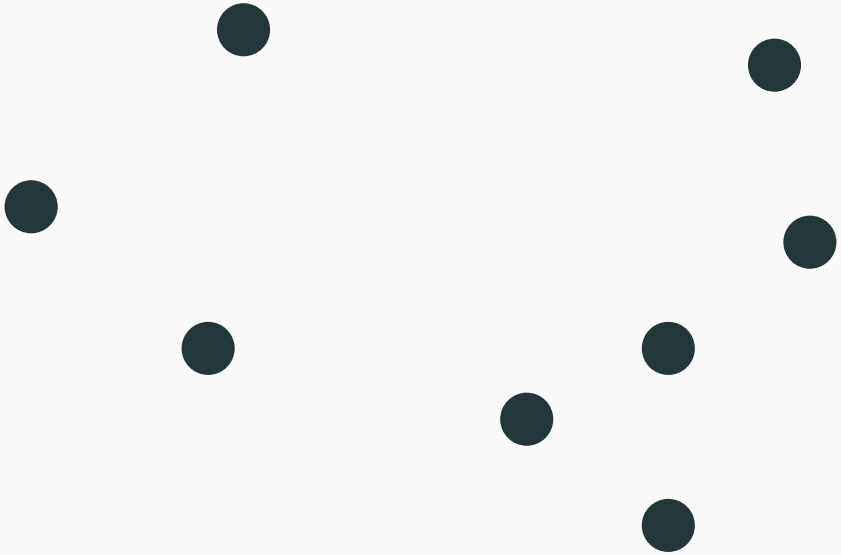

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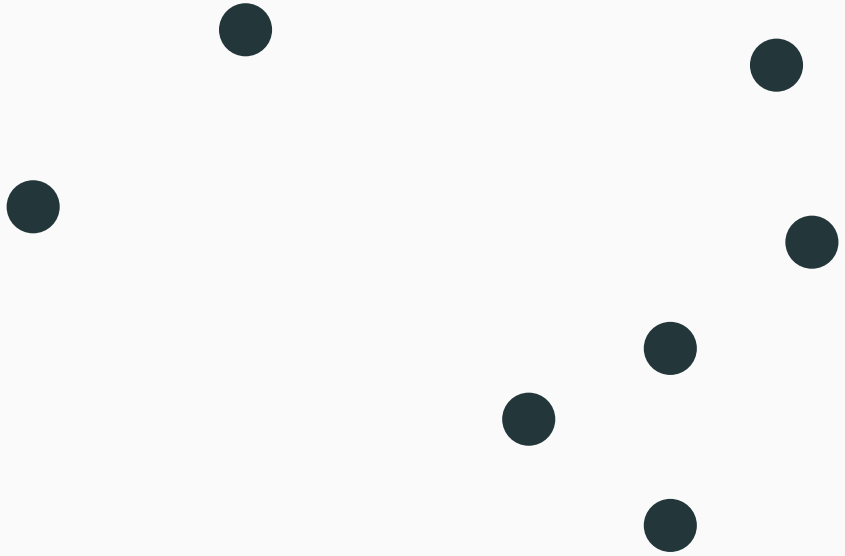
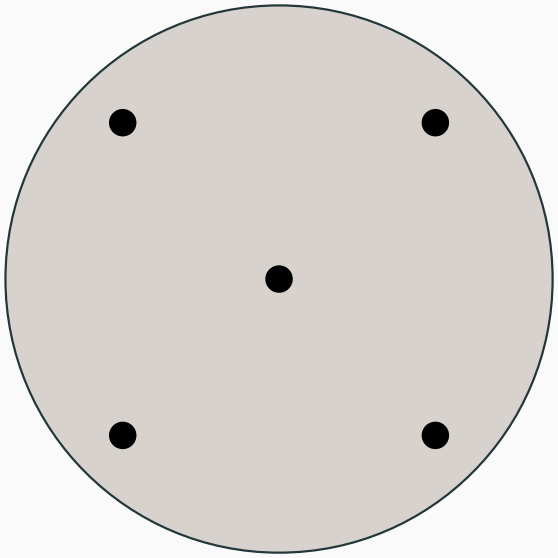
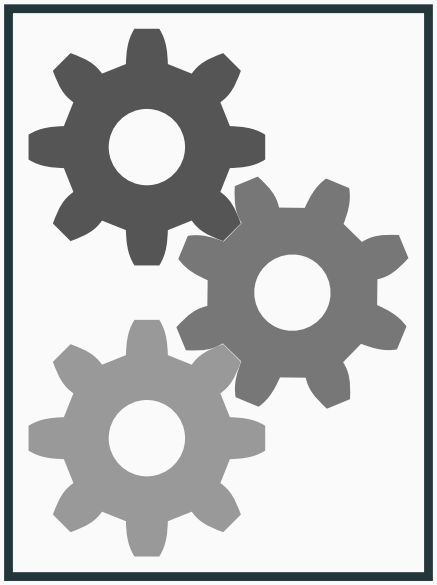


Dynamic Setting

Dynamic point set P



Dynamic algorithm



k-Center solution

$S = \{c_1, \ldots, c_k\}$

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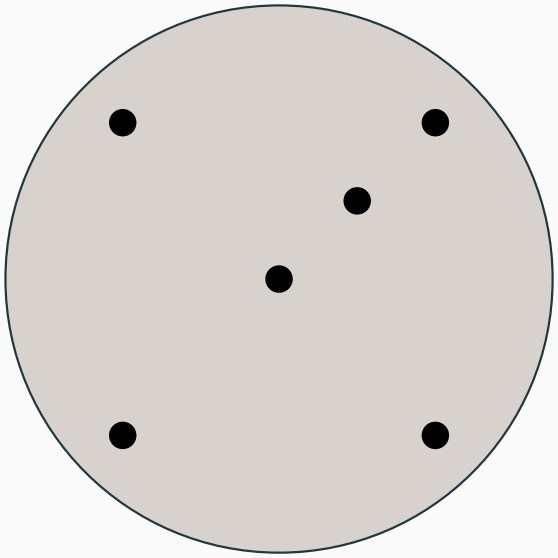
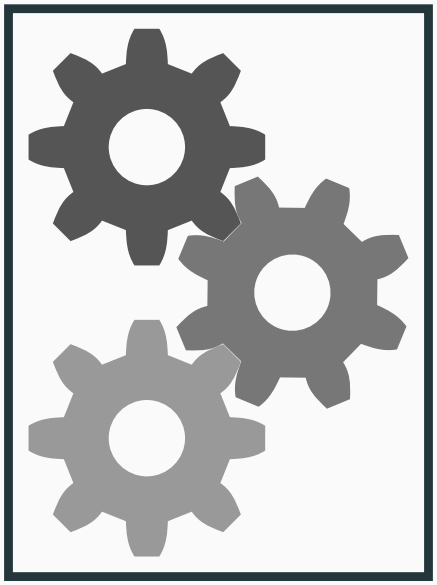


Adversary inserts new point

Dynamic algorithm



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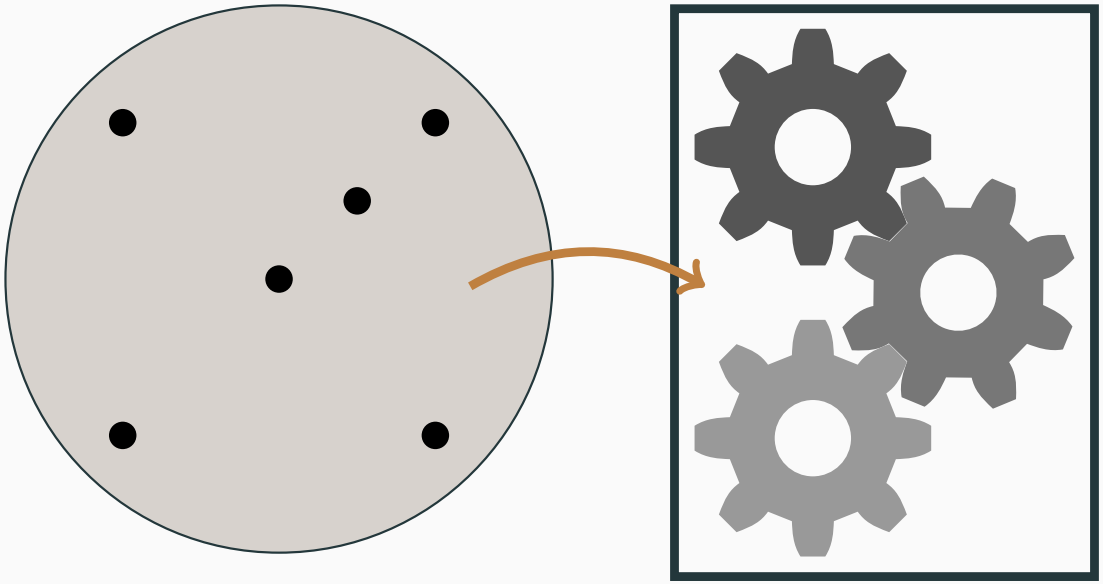

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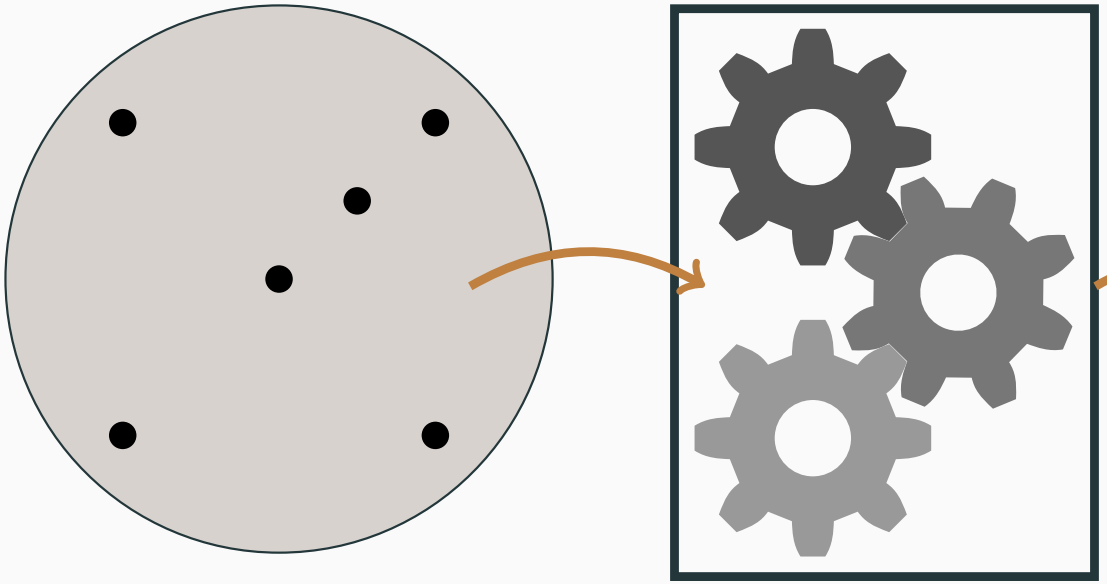

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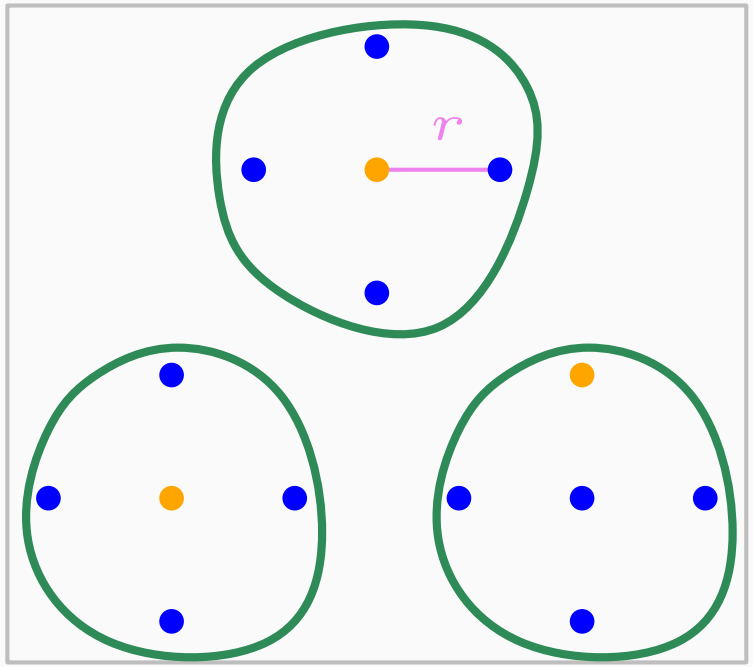
Algorithm updates the k centers

Dynamic k-Center Clustering

Fully Dynamic k-Center

Updates: Insert/Delete point

$k = 3$ r is the cost



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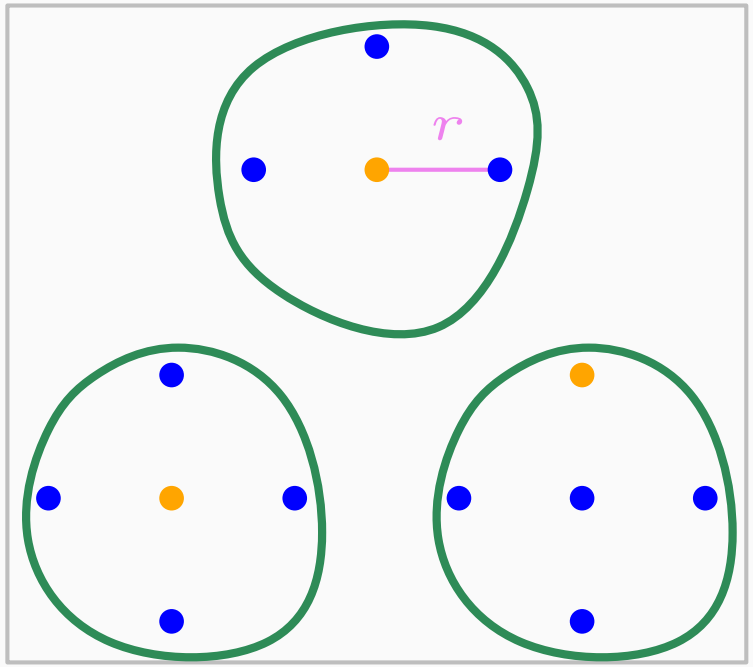

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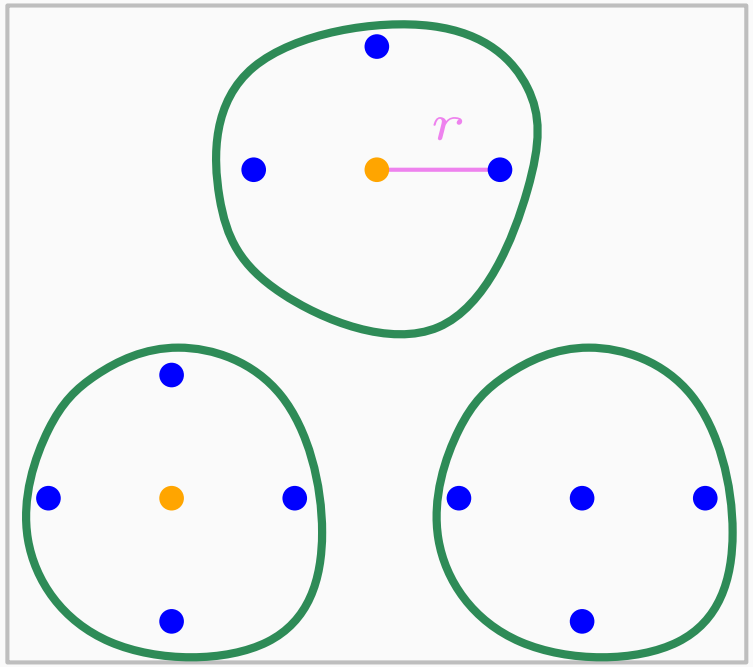

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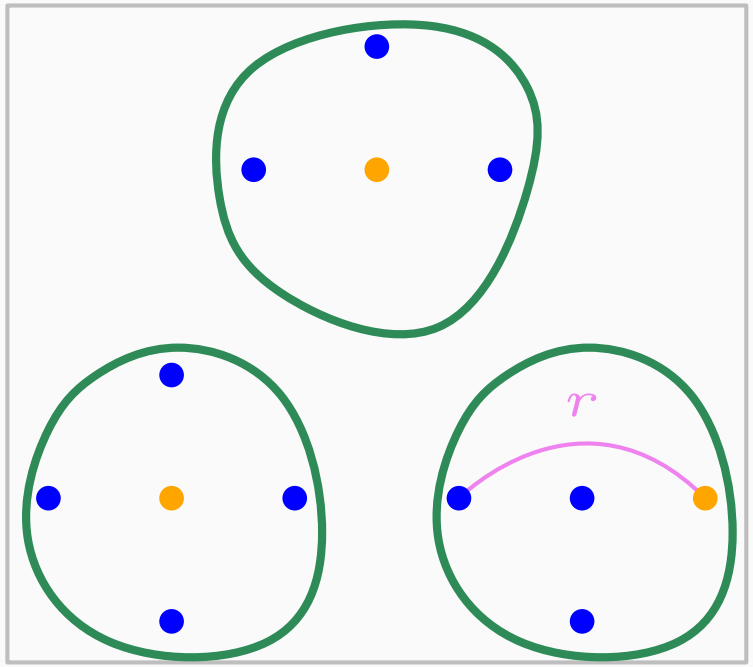

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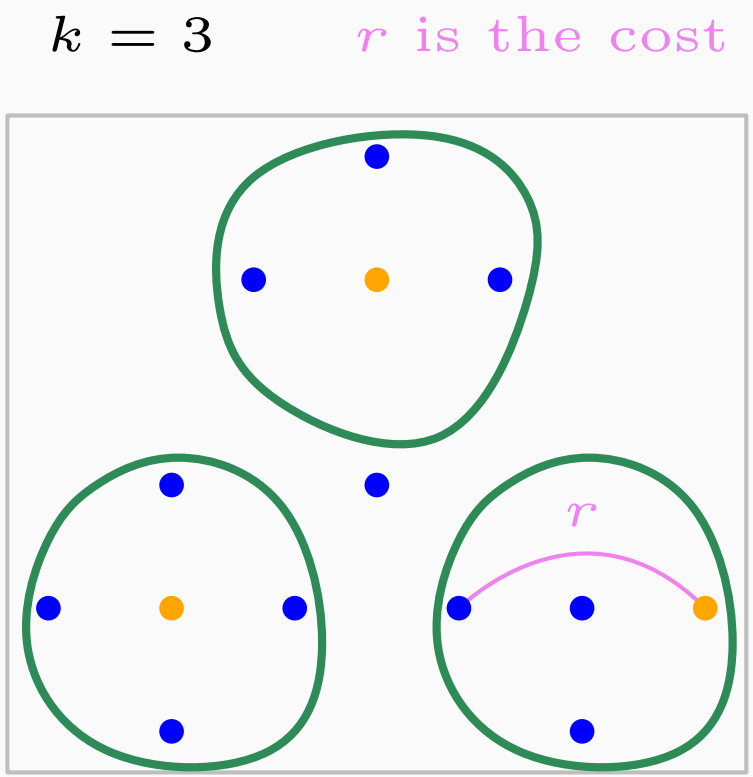

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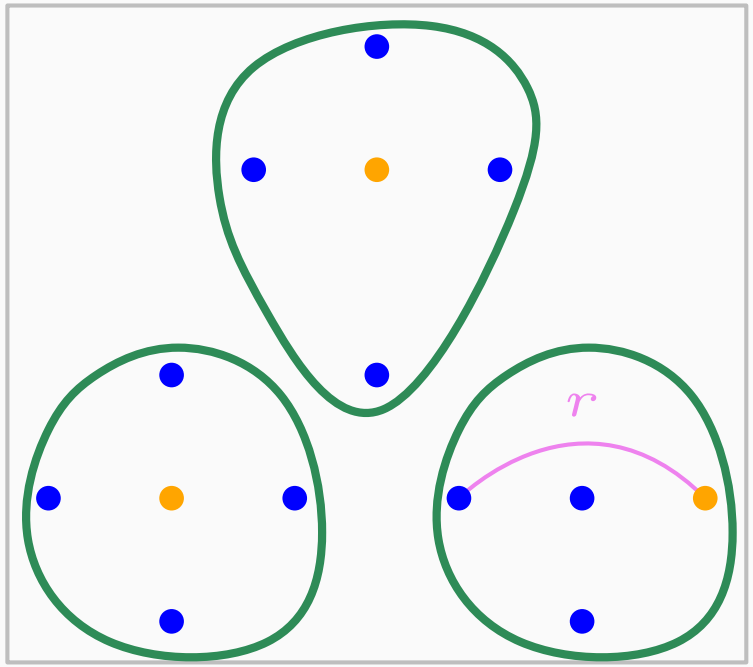

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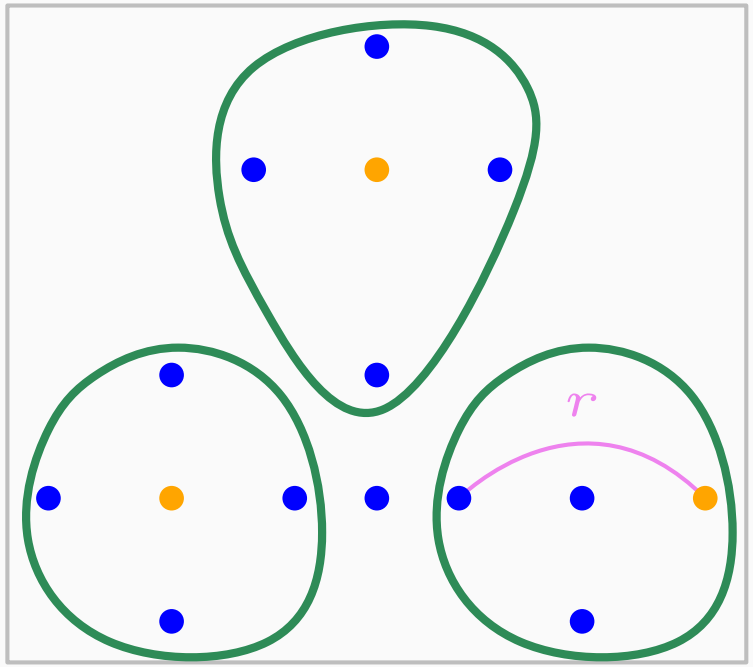

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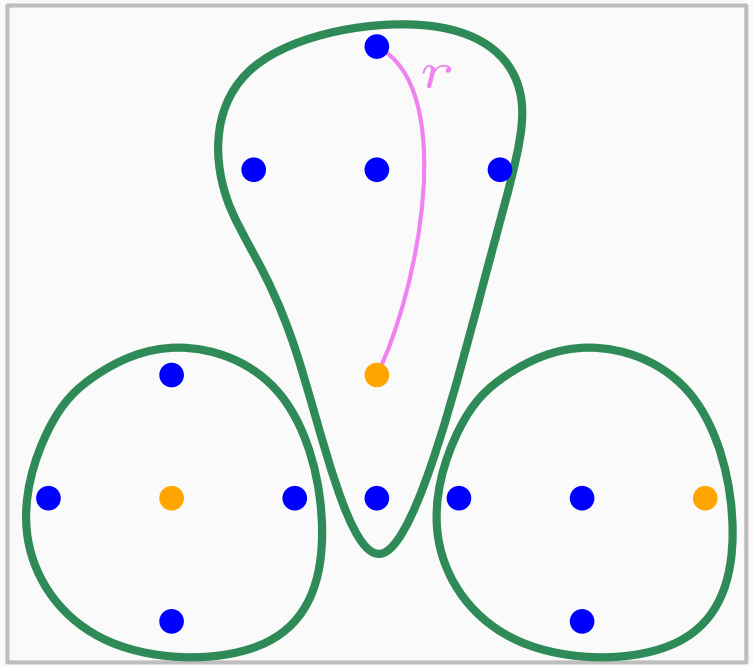

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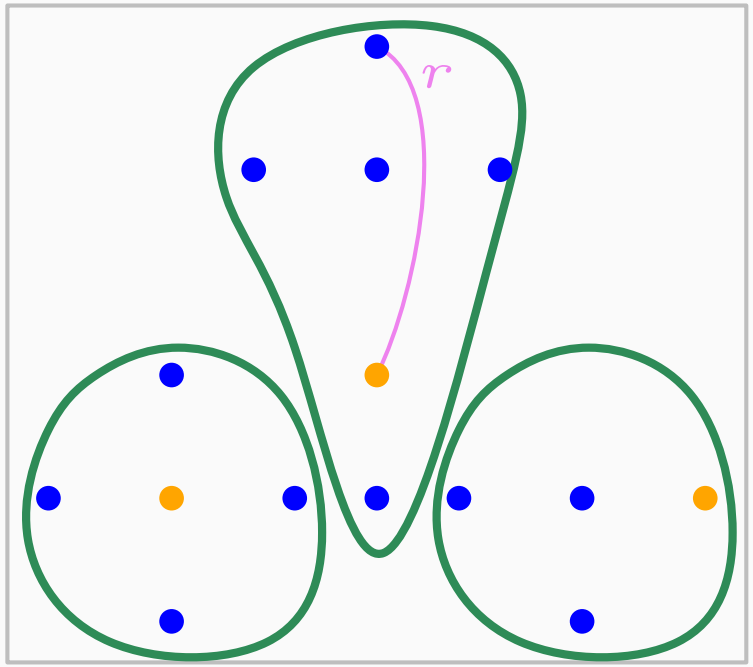

Updates: Insert/Delete point

Optimize:

$O(1)$ approximation

$O(1)$ recourse

Recourse: the number of changes to centers

Fully Dynamic k-Center

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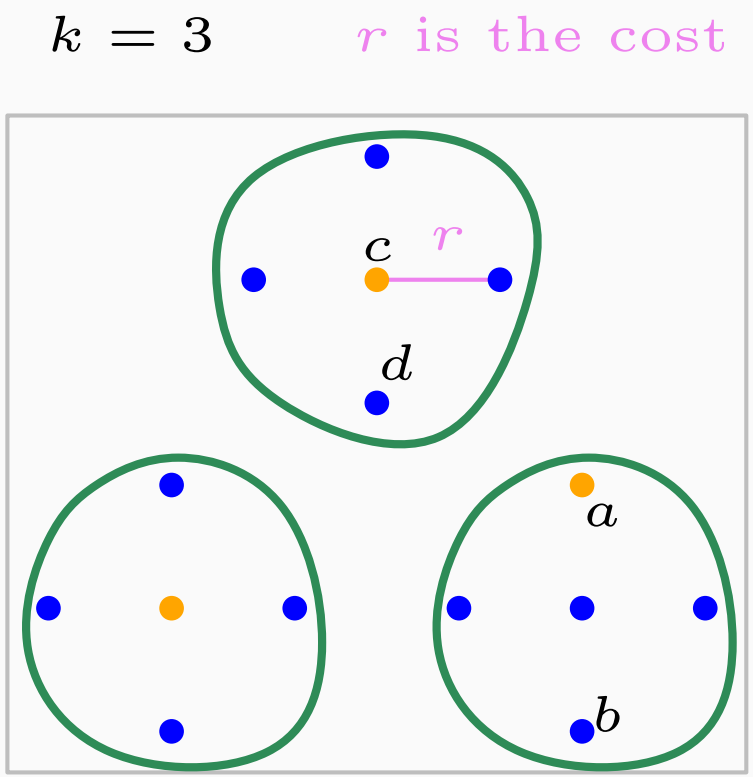

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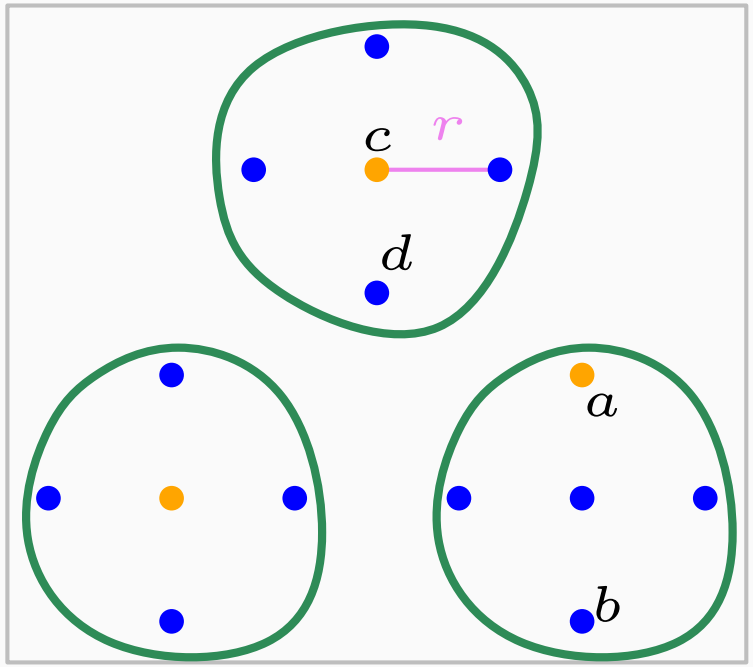

Adversary deletes a

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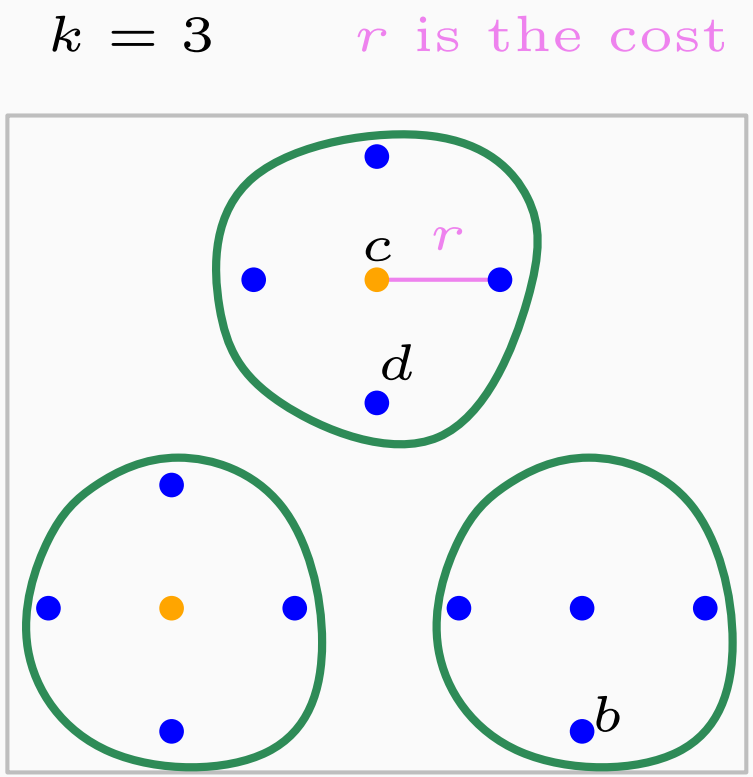

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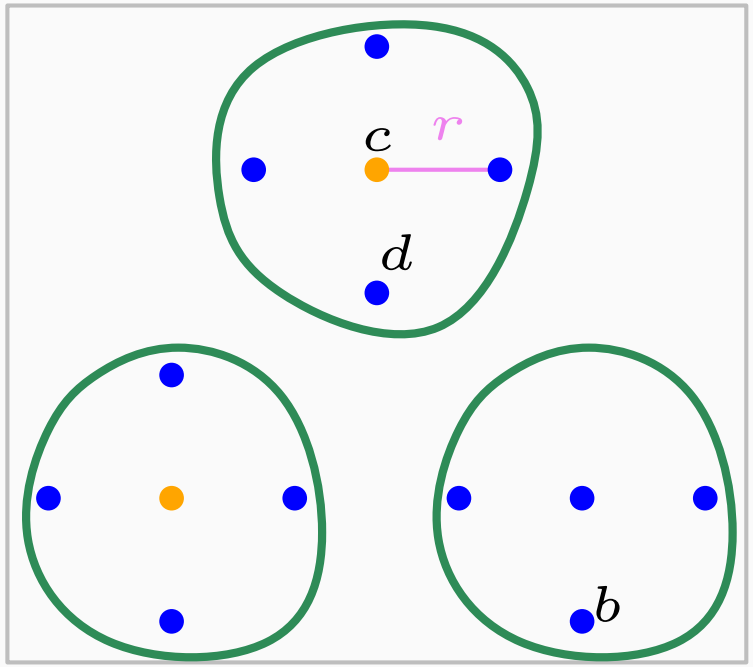

Remove a from centers

Add b to centers

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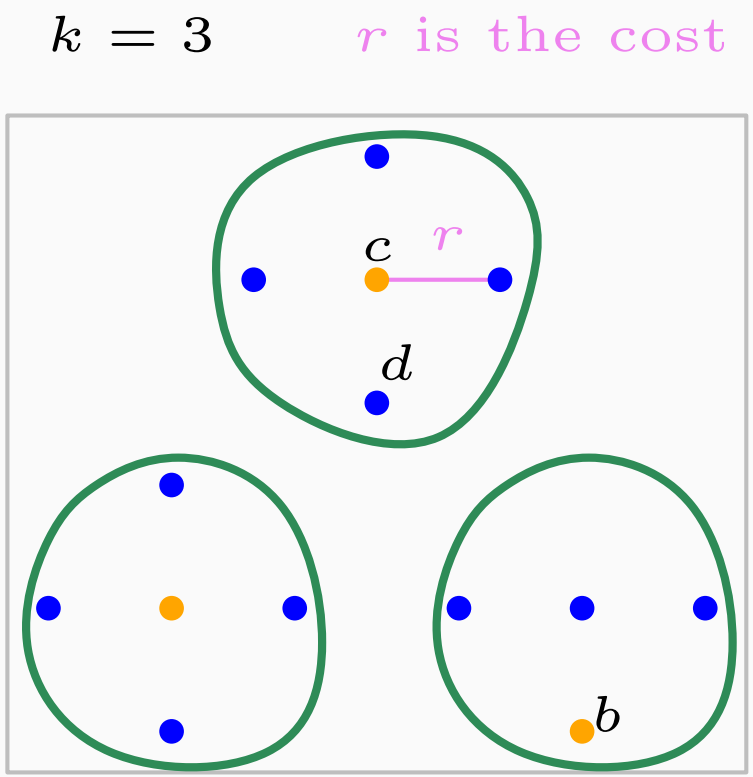

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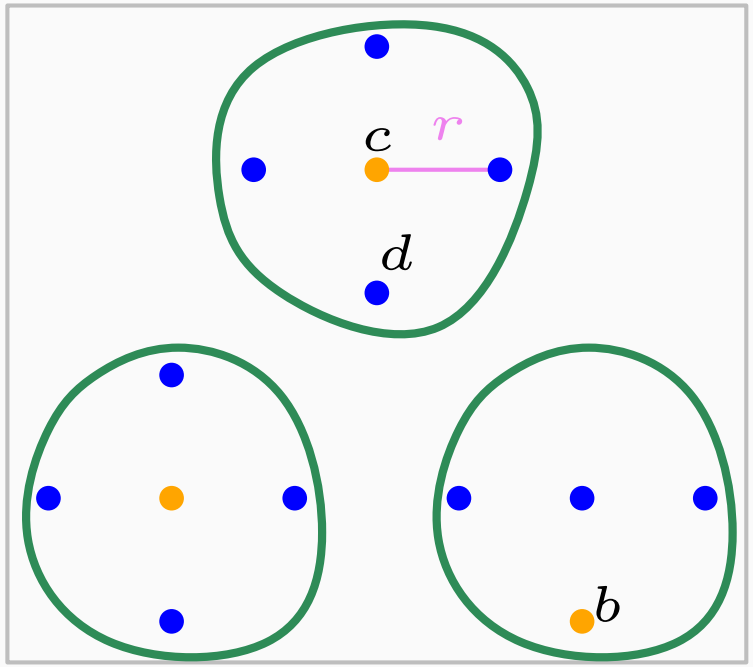

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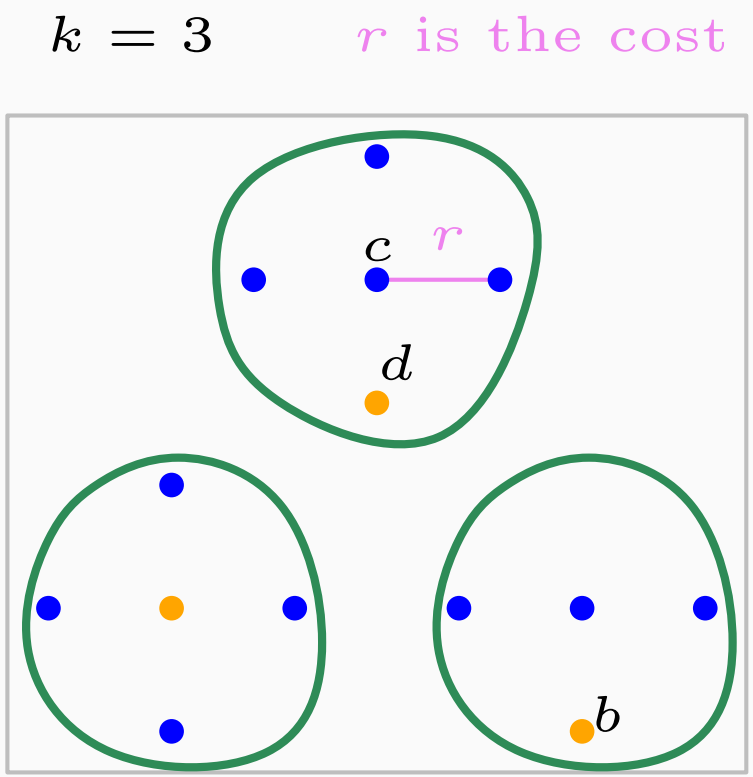

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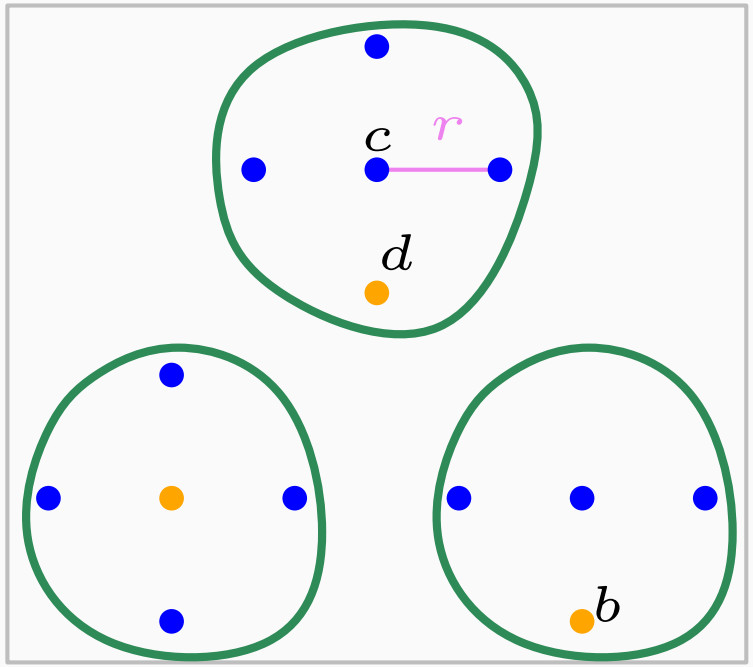

recourse is 2

Recourse: the number of changes to centers

Setup for k-Center Algorithm

Certificate for k-Center Clustering

Lemma ([Hochbaum, Shmoys '86])

If $k + 1$ *points in* P *whose pairwise distances* $\geq r$

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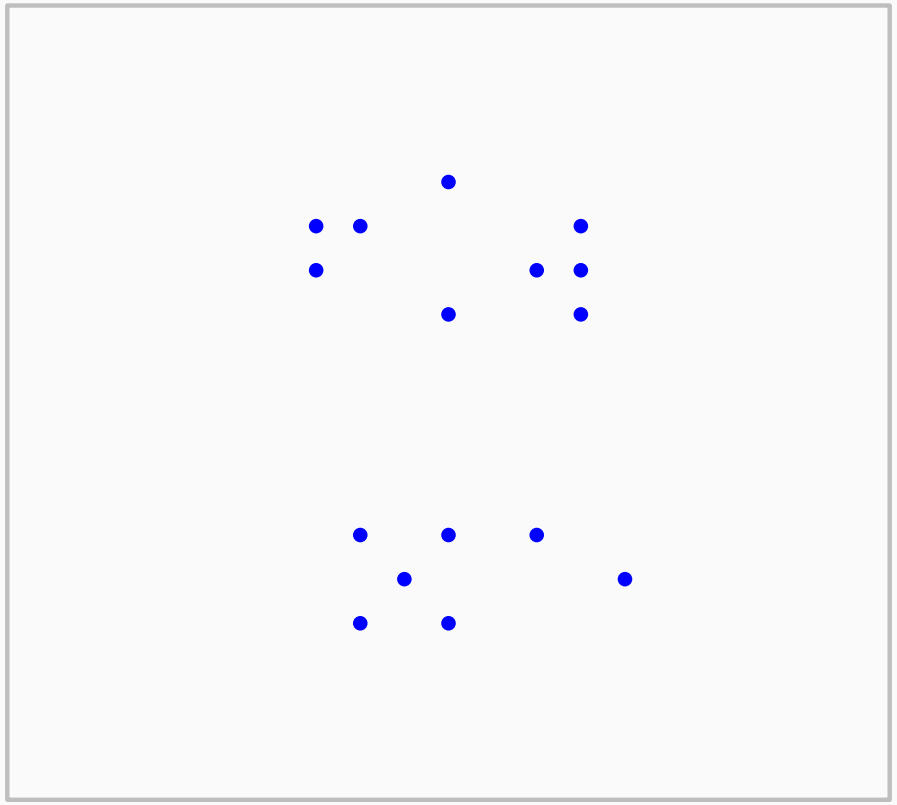

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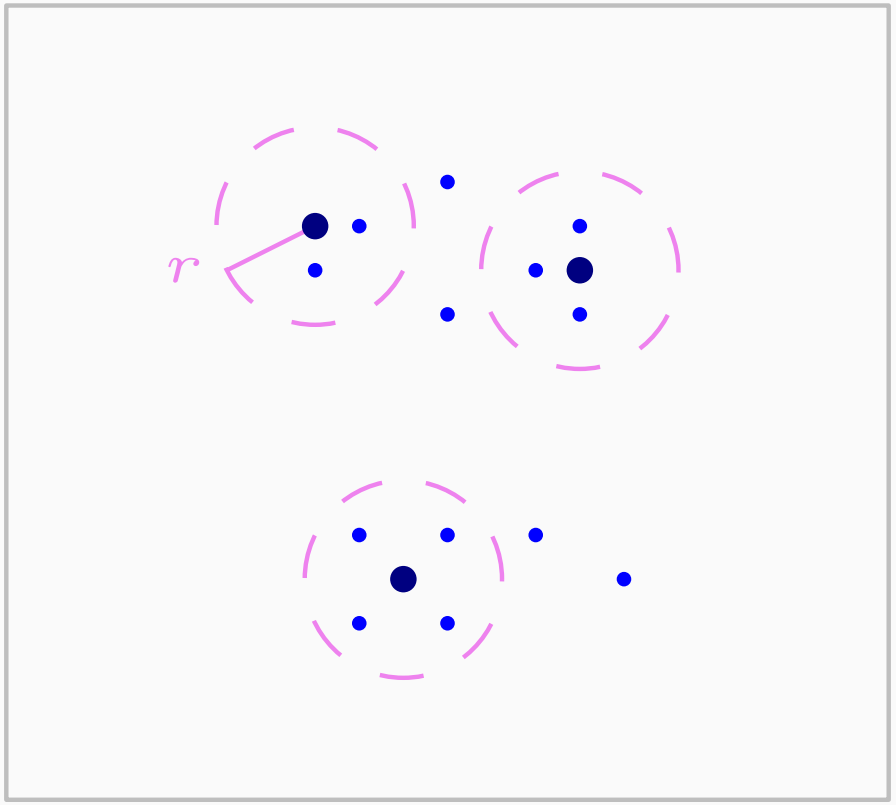

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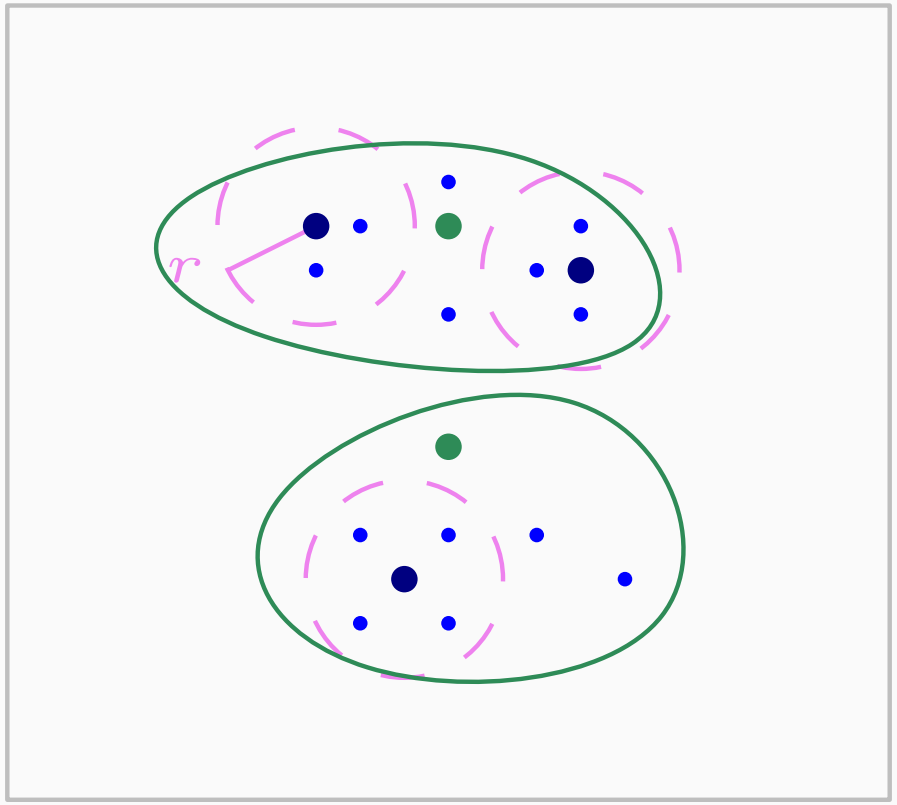

Approximation Ratio

Invariants for fixed constants $\alpha \geq 1$

Invariant 1: $|S| = k$

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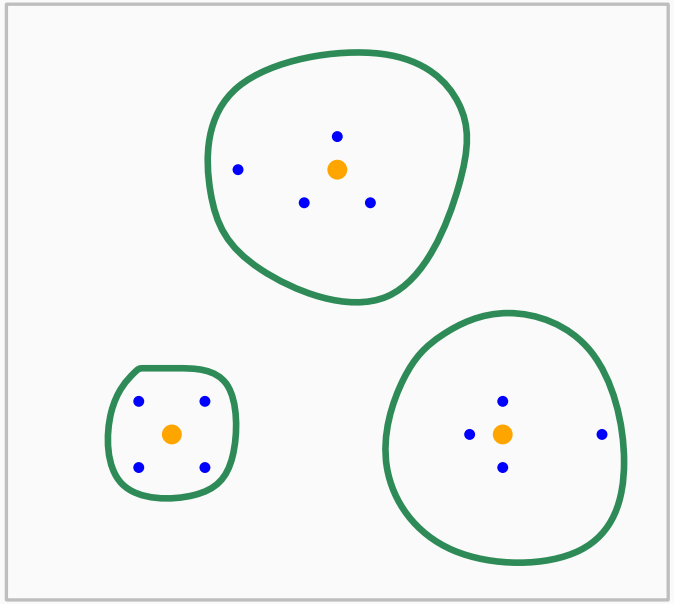

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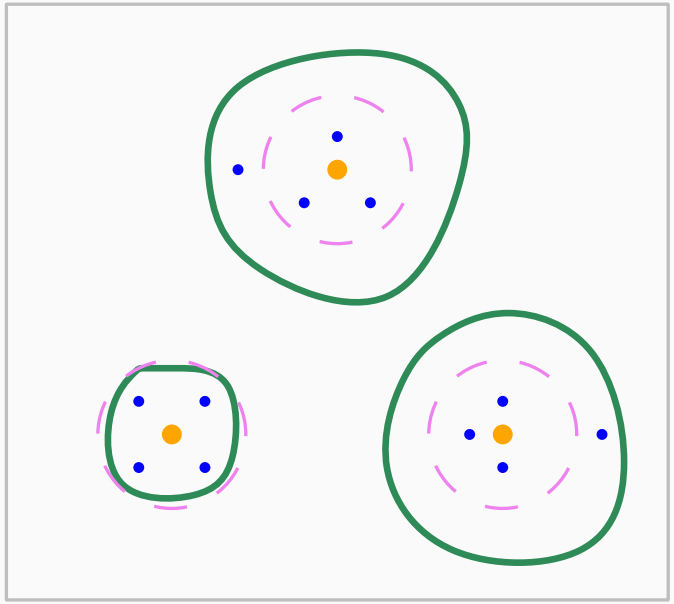

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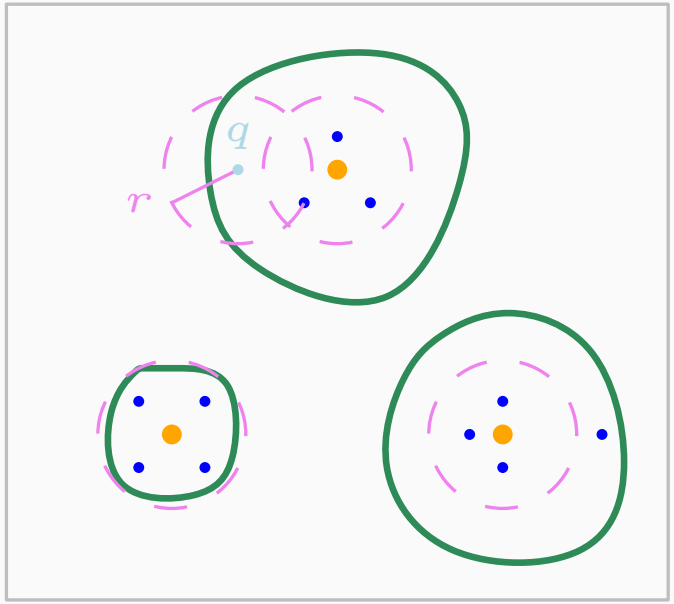
$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

$k = 3$



Approximation Ratio

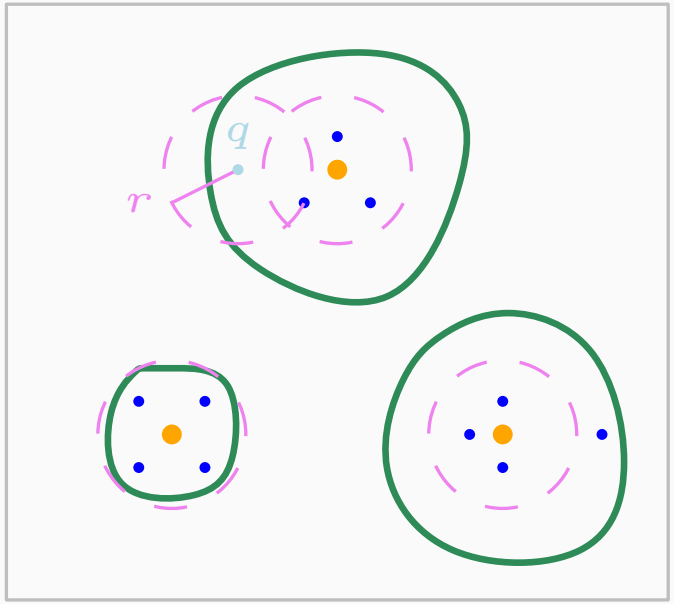
Invariants for fixed constants $\alpha \geq 1$

Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \text{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \text{dist}(q, S) \geq r$

$k = 3$



Invariant 1 and Invariant 2

$r \leq 2 \cdot OPT$

Approximation Ratio

Invariants for fixed constants $\alpha \geq 1$

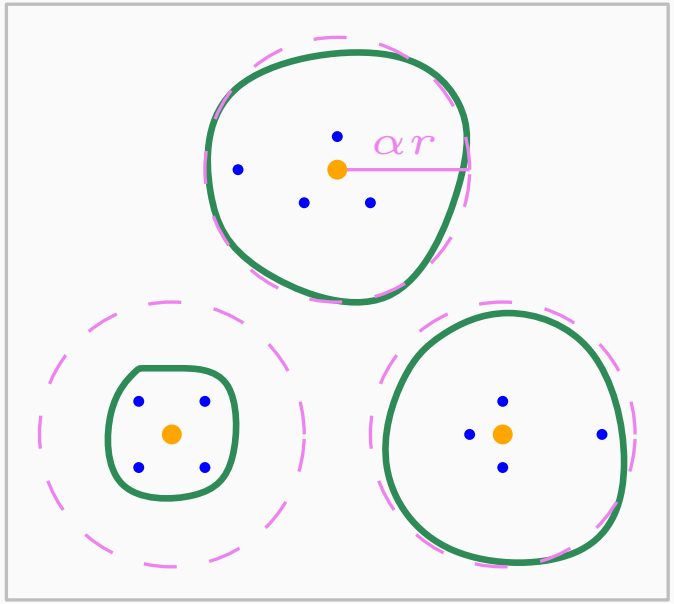
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \text{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \text{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \text{dist}(p, S) \leq \alpha \cdot r$

$k = 3$



Invariant 1 and Invariant 2

$r \leq 2 \cdot OPT$

Approximation Ratio

Invariants for fixed constants $\alpha \geq 1$

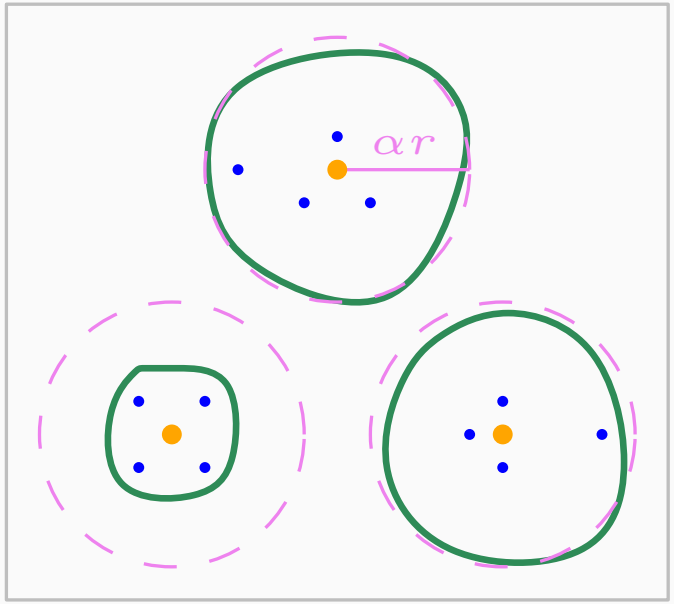
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq \alpha \cdot r$

$k = 3$



Invariant 1 and Invariant 2

$r \leq 2 \cdot OPT$

Together with Invariant 3

S is (2α)-approximate

Incremental Setting

Point Insertion

Invariants

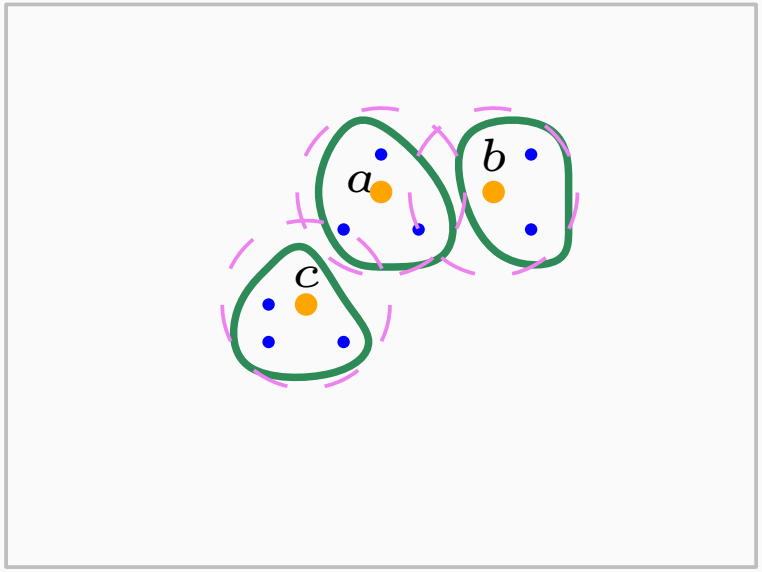
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



Point Insertion

Invariants

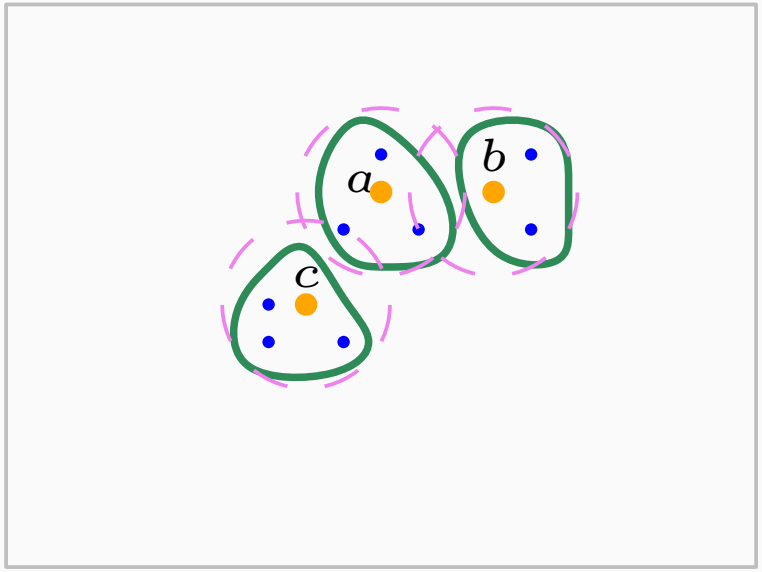
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



Observation: After a point insertion, Invariant 3 could be violated

Point Insertion

Invariants

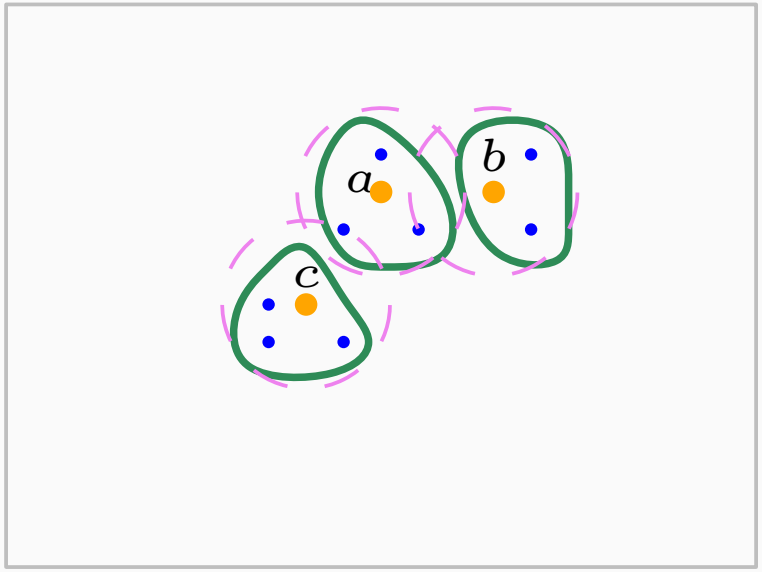
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

Point Insertion

Invariants

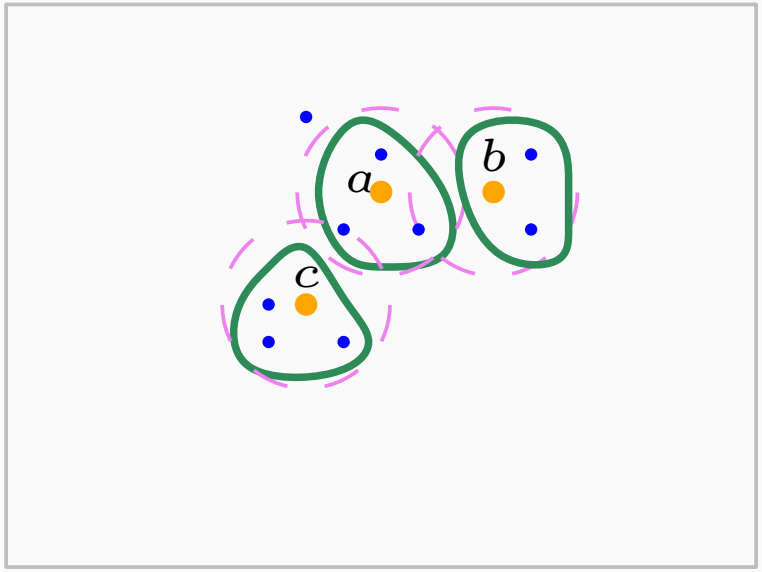
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) > r$

Point Insertion

Invariants

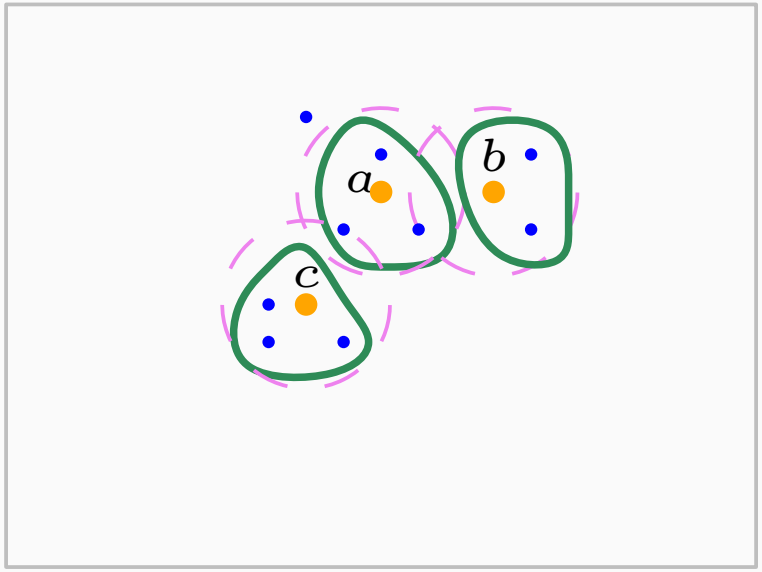
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) > r$

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

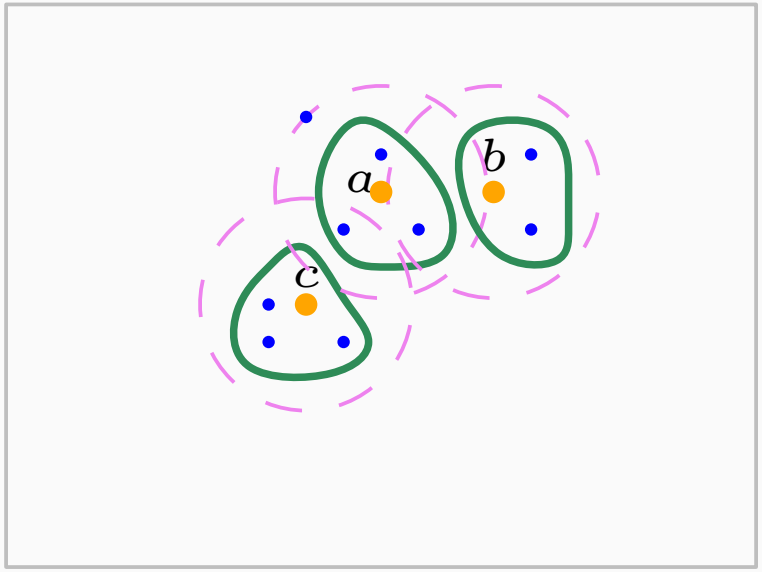
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\frac{r}{2} < \mathrm{dist}(p^+, S) \leq r$

Double r

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

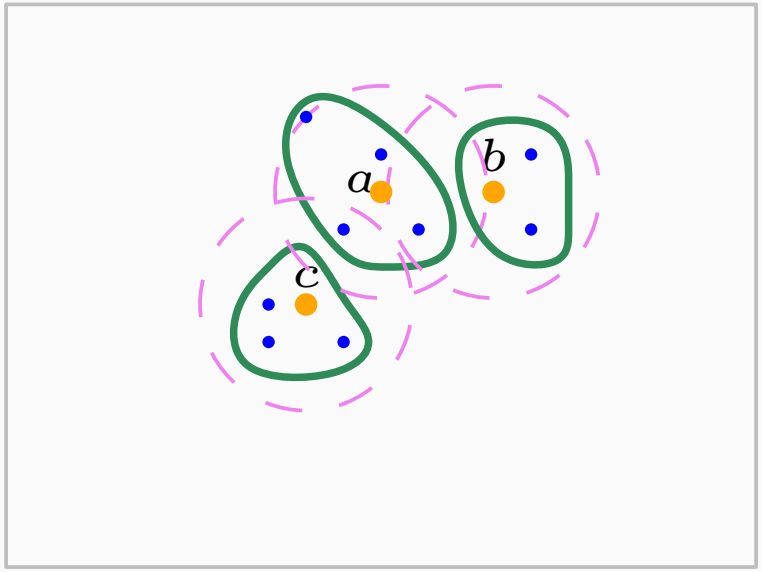
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq r$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq r$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\frac{r}{2} < \mathrm{dist}(p^+, S) \leq r$

Double r

Add p^+ to C_a

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

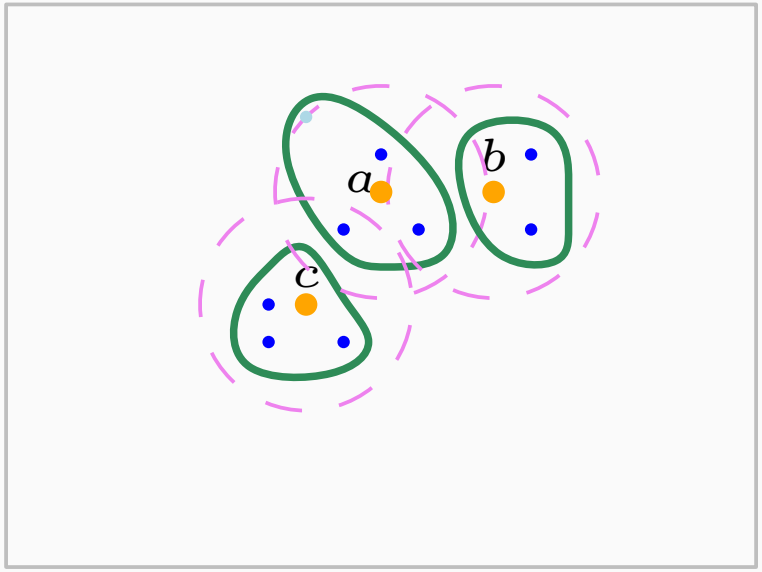
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\frac{r}{2} < \mathrm{dist}(p^+, S) \leq r$

Double r

Add p^+ to C_a, $q = p^+$

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

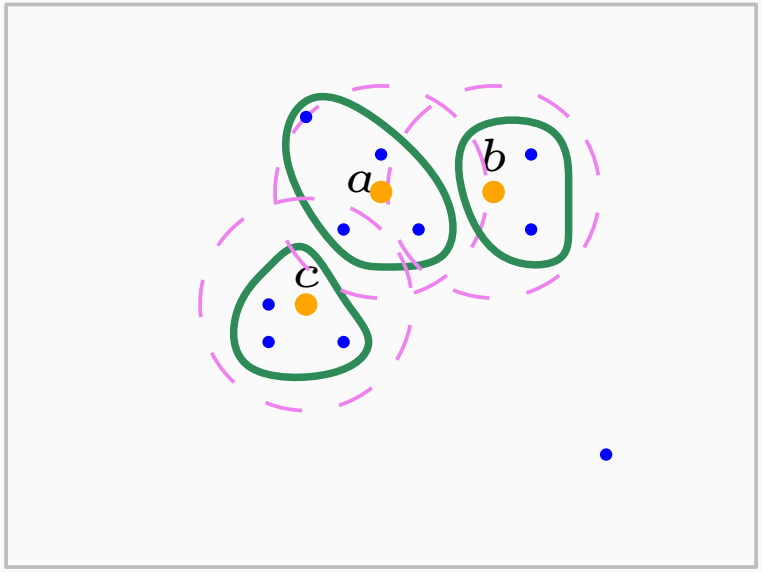
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) > r$

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

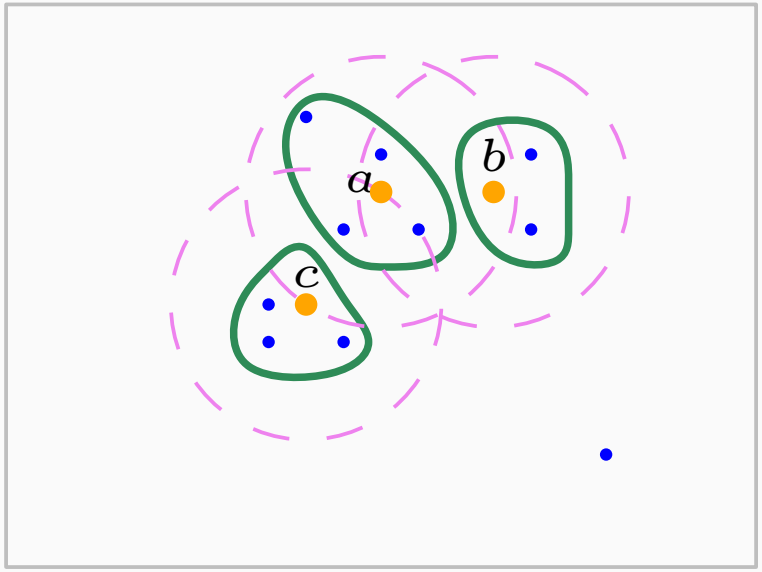
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq \frac{r}{2}$

$\mathrm{dist}(p^+, S) > r$

Double r

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

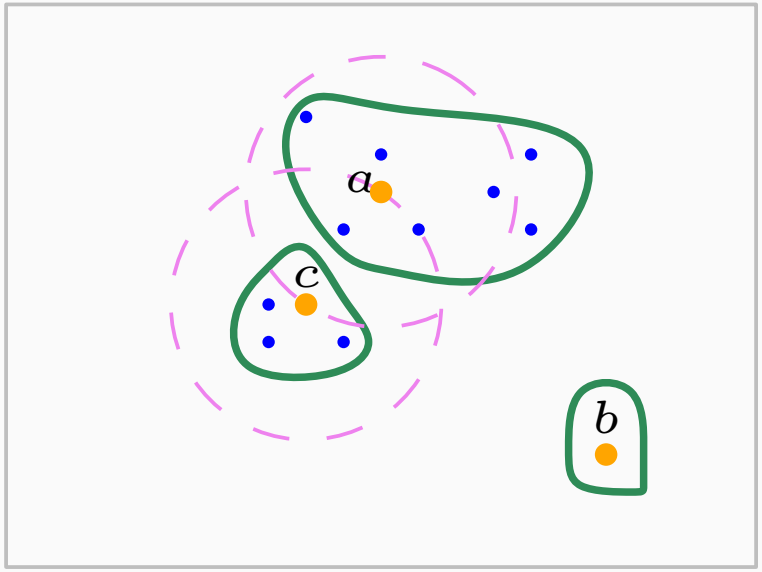
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq 2r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) \leq r$

Double r

Add C_b to C_a, C_a becomes *extended*

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

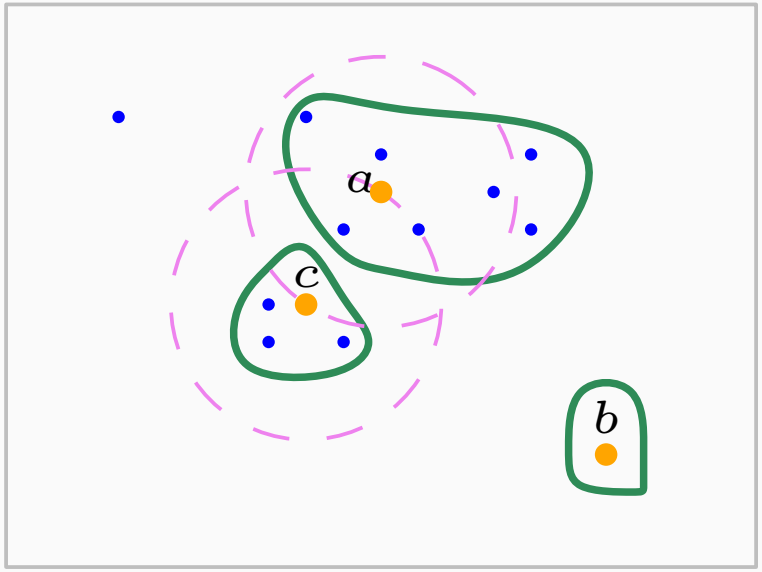
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq 2r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) > r$

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

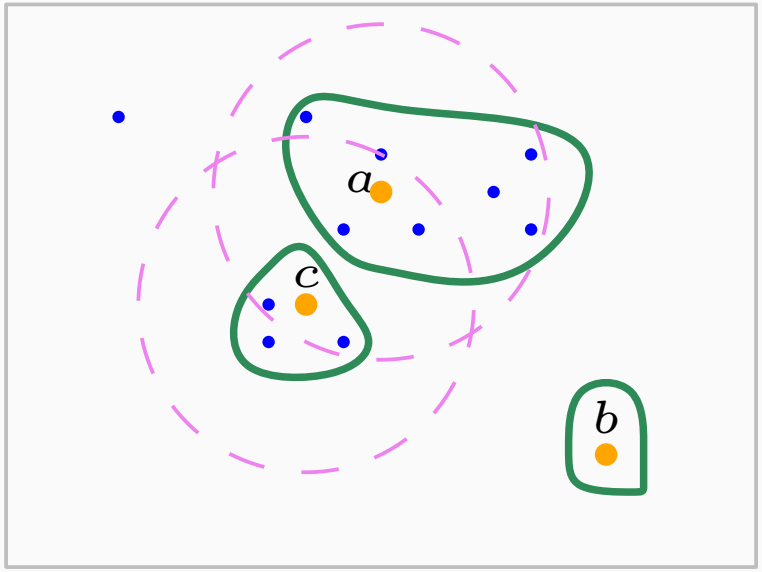
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq 2r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq \frac{r}{2}$

$\mathrm{dist}(p^+, S) > r$

Double r

Double r until **Inv 3** is satisfied or **Inv 2** is tight

Point Insertion

Invariants

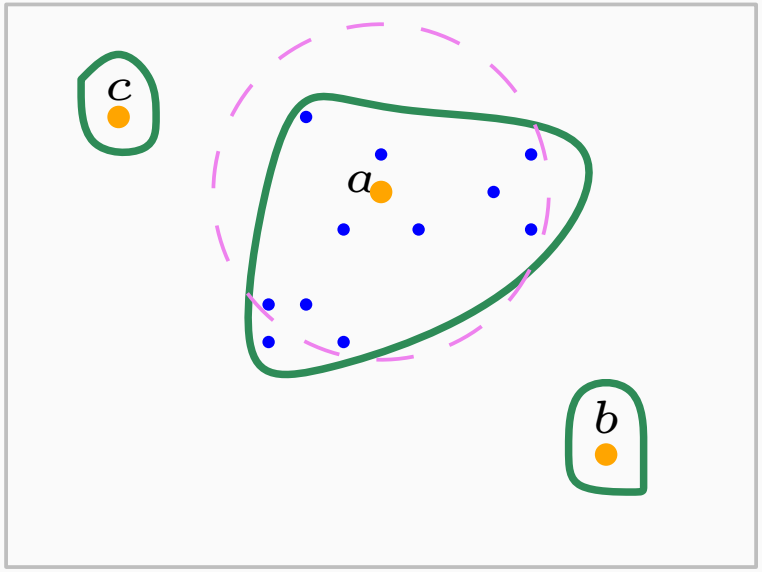
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq 2r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) \leq r$

Double r

Point Insertion

Invariants

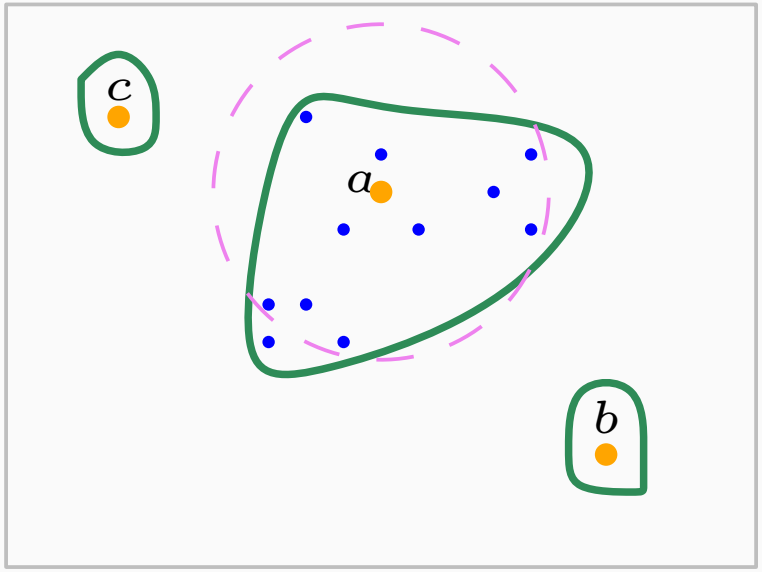
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq 2r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) \leq r$

Double r

Concern: Can the approximation ratio explode?

Point Insertion

Invariants

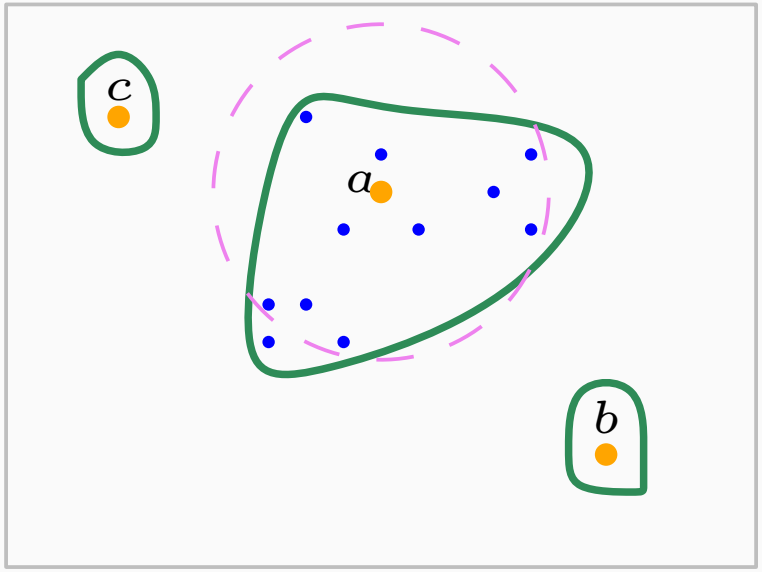
Invariant 1: $|S| = k$

Invariant 2: $\forall q_1, q_2 \in S : \ \mathrm{dist}(q_1, q_2) \geq \frac{r}{2}$

$\exists q \in P \setminus S : \ \mathrm{dist}(q, S) \geq \frac{r}{2}$

Invariant 3: $\forall p \in P : \ \mathrm{dist}(p, S) \leq 2r$

$k = 3$



$\mathrm{dist}(a, b), \mathrm{dist}(a, c), \mathrm{dist}(b, c) \geq r$

$\mathrm{dist}(p^+, S) \leq r$

Double r

Order using maximal distance-r independent set on centers